

Supplementary Data for “SEISMIC-RNA: scalable, fully automated RNA structure ensemble analysis with bias correction”

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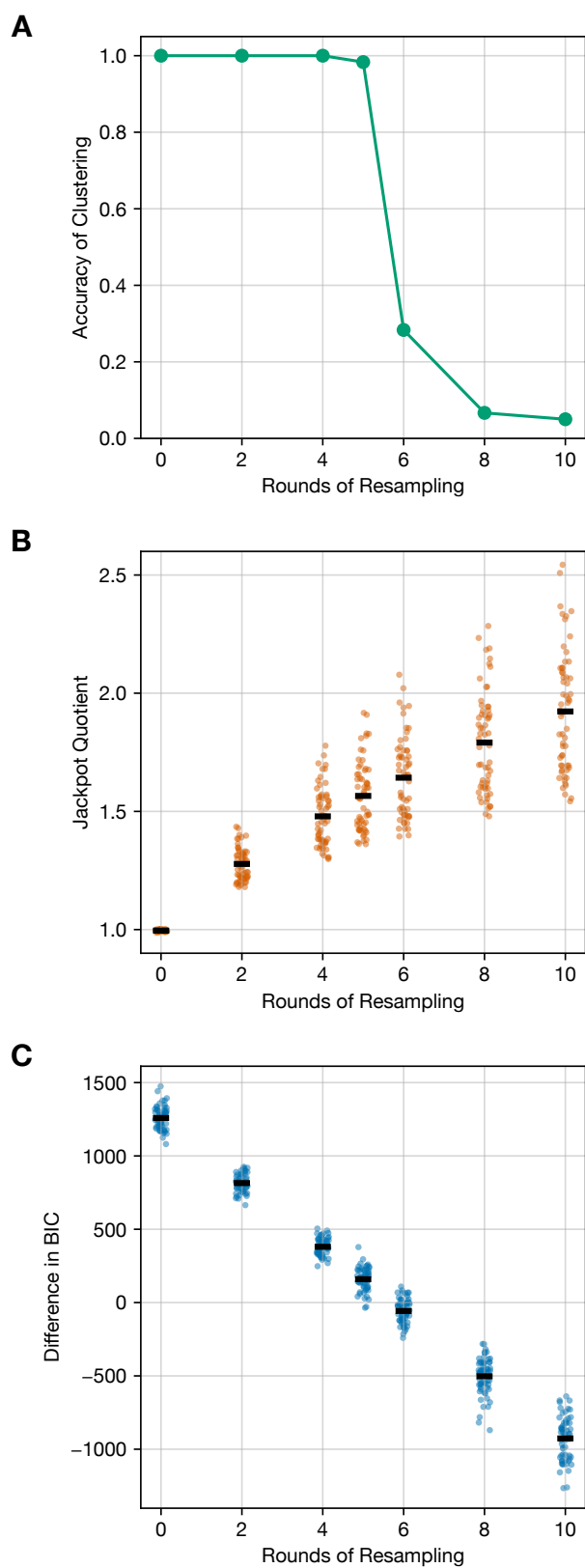
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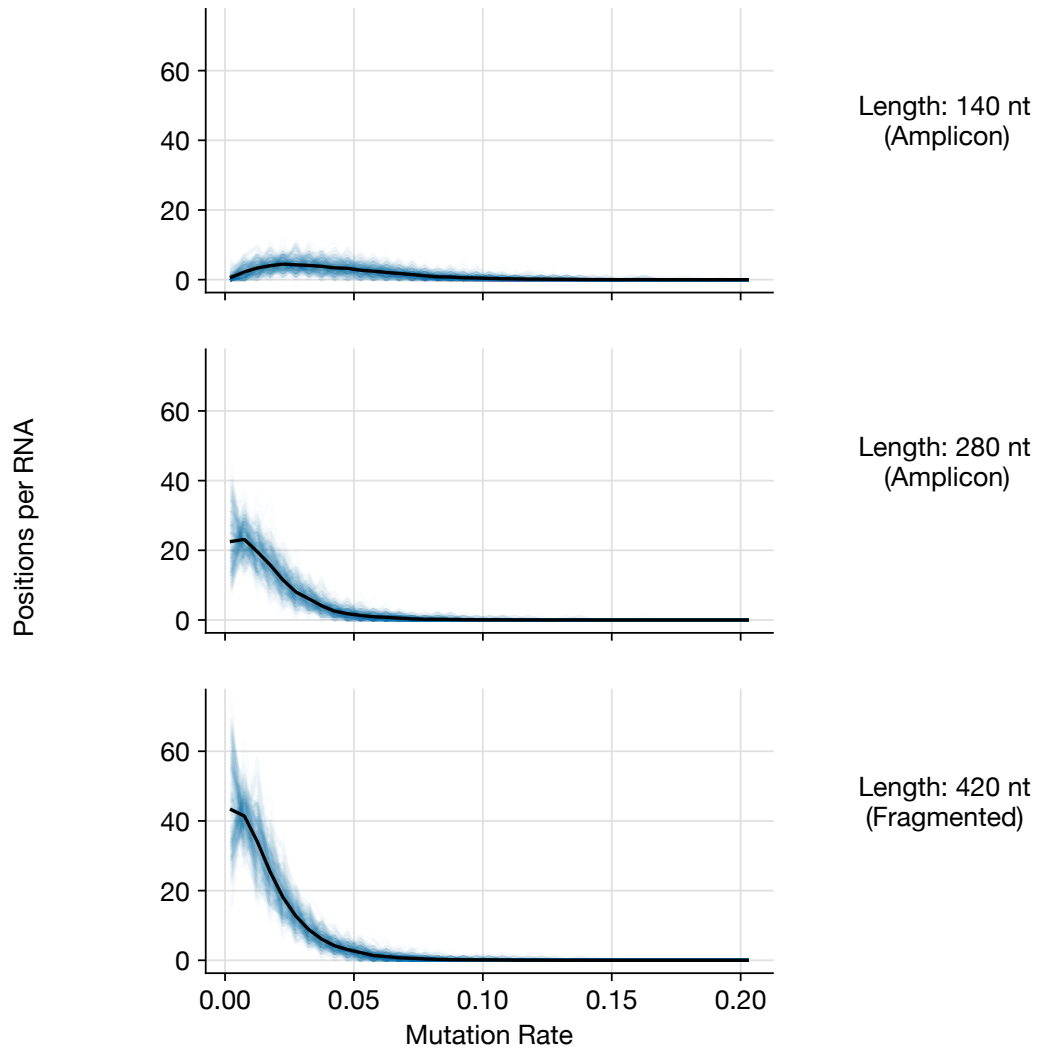
Authors’ note: We originally described the topic of this manuscript, SEISMIC-RNA, in another preprint [1]. That original preprint introduced two methods: SEISMIC-RNA and SEARCH-MaP. After posting it, we realized that we would need to split it into two manuscripts to have space to describe each method with sufficient detail. This manuscript is a revision of only the portion about SEISMIC-RNA – a substantial change in focus that we think warrants posting it as a new manuscript, rather than as a new version of the original manuscript. Nevertheless, as a revision, this manuscript reproduces some text, figures, equations, and other material from the original [1].

Supplementary Figures

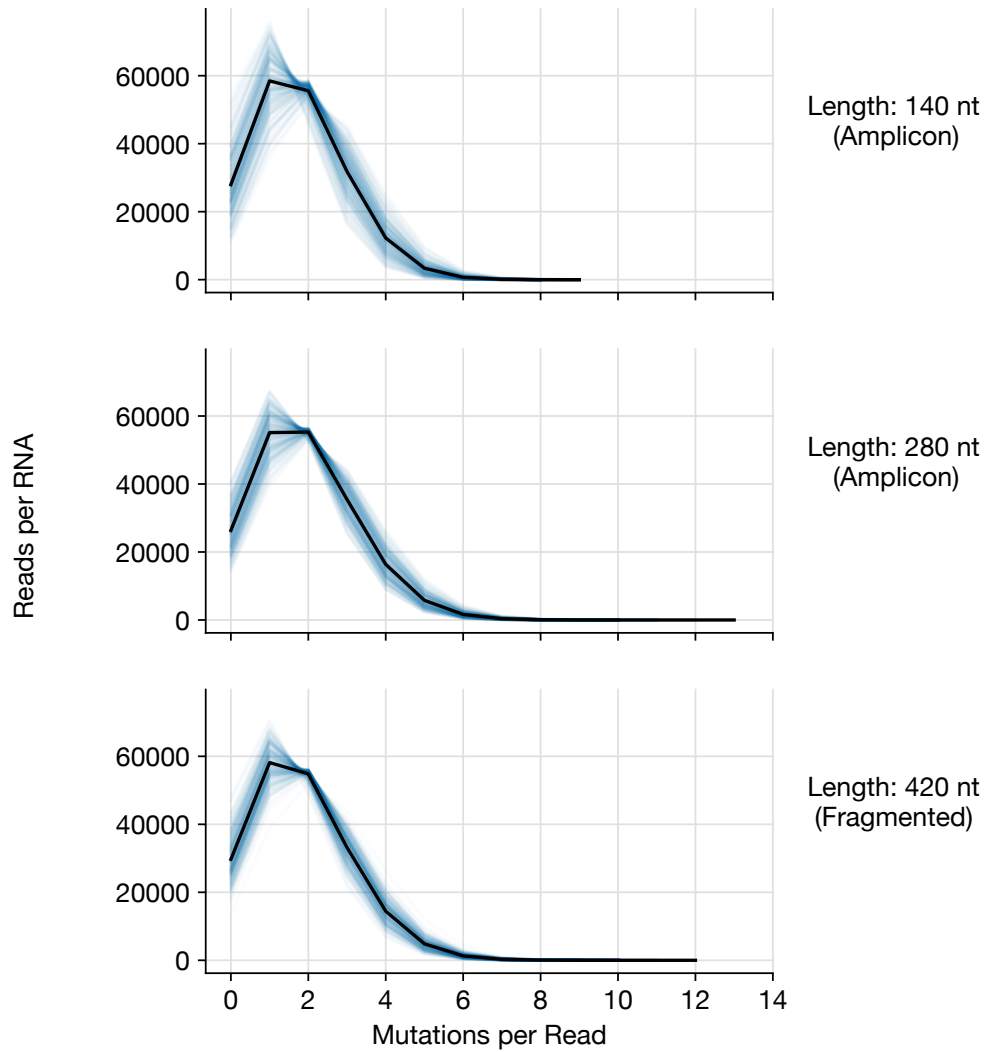


Supplementary Figure 1: **Jackpotting causes false clusters.** (Continued on next page.)

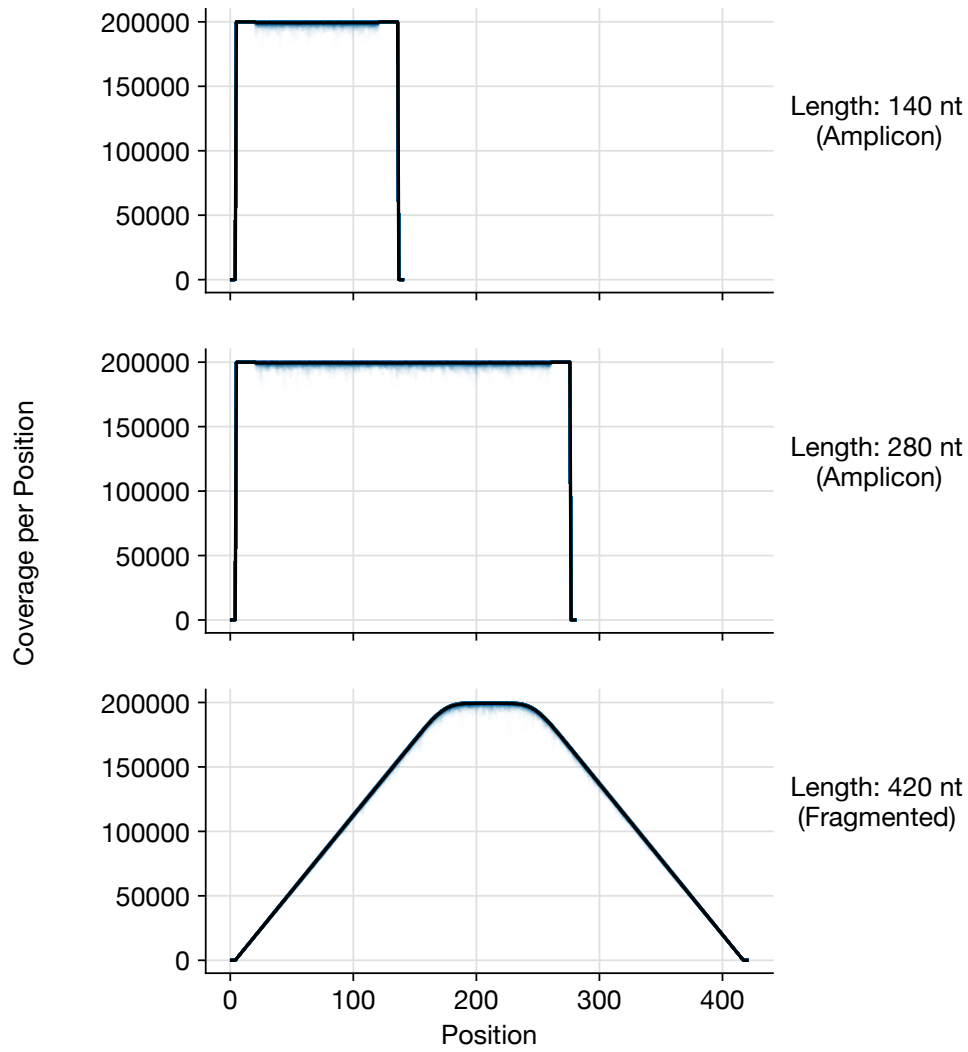
Supplementary Figure 1: (Continued from previous page.) **(A)** The fraction of 60 FASTQ files (each of a random 280 nt RNA with 1 structure and 200,000 simulated reads) for which SEISMIC-RNA correctly found 1 cluster as a function of the number of rounds of resampling 200,000 reads with replacement. **(B)** The distribution of jackpotting quotients for each number of rounds of resampling. Horizontal black bars show means. **(C)** The distribution of the difference in BIC between 2 clusters and 1 cluster for each number of rounds of resampling. Horizontal black bars show means.



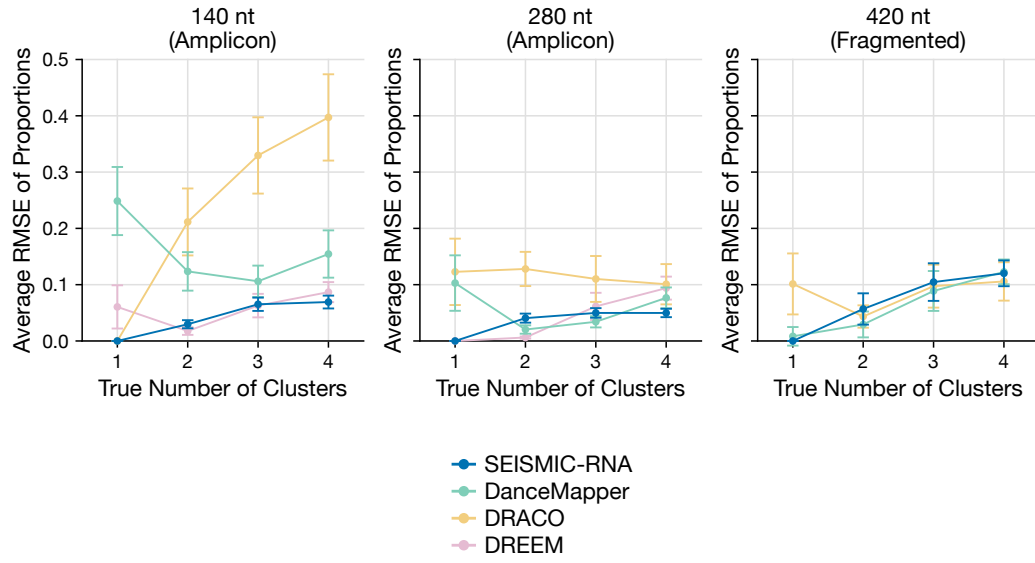
Supplementary Figure 2: **Histogram of mutation rates using 200,000 reads.** For each reference length, the histogram of the mutation rates on A and C positions for all 240 RNAs with 200,000 simulated reads (transparent blue). The black line shows the average. The bin size for the histogram is 0.005.



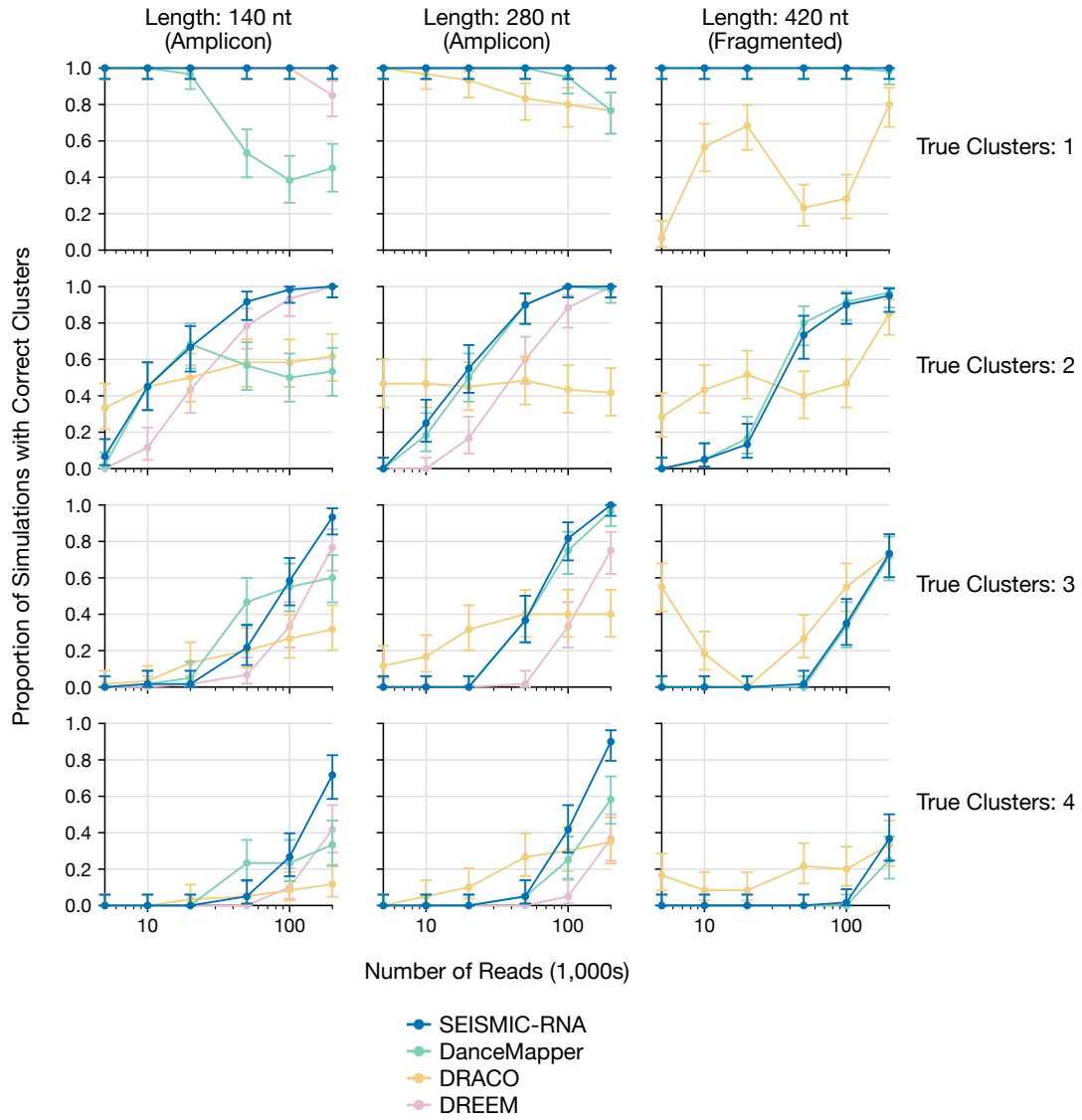
Supplementary Figure 3: **Histogram of number of mutated bases per read using 200,000 reads.** For each reference length, the histogram of the number of mutated bases for all 240 RNAs with 200,000 simulated reads (transparent blue). The black line shows the average. The RNAs were simulated to have an average of 2 mutations per read.



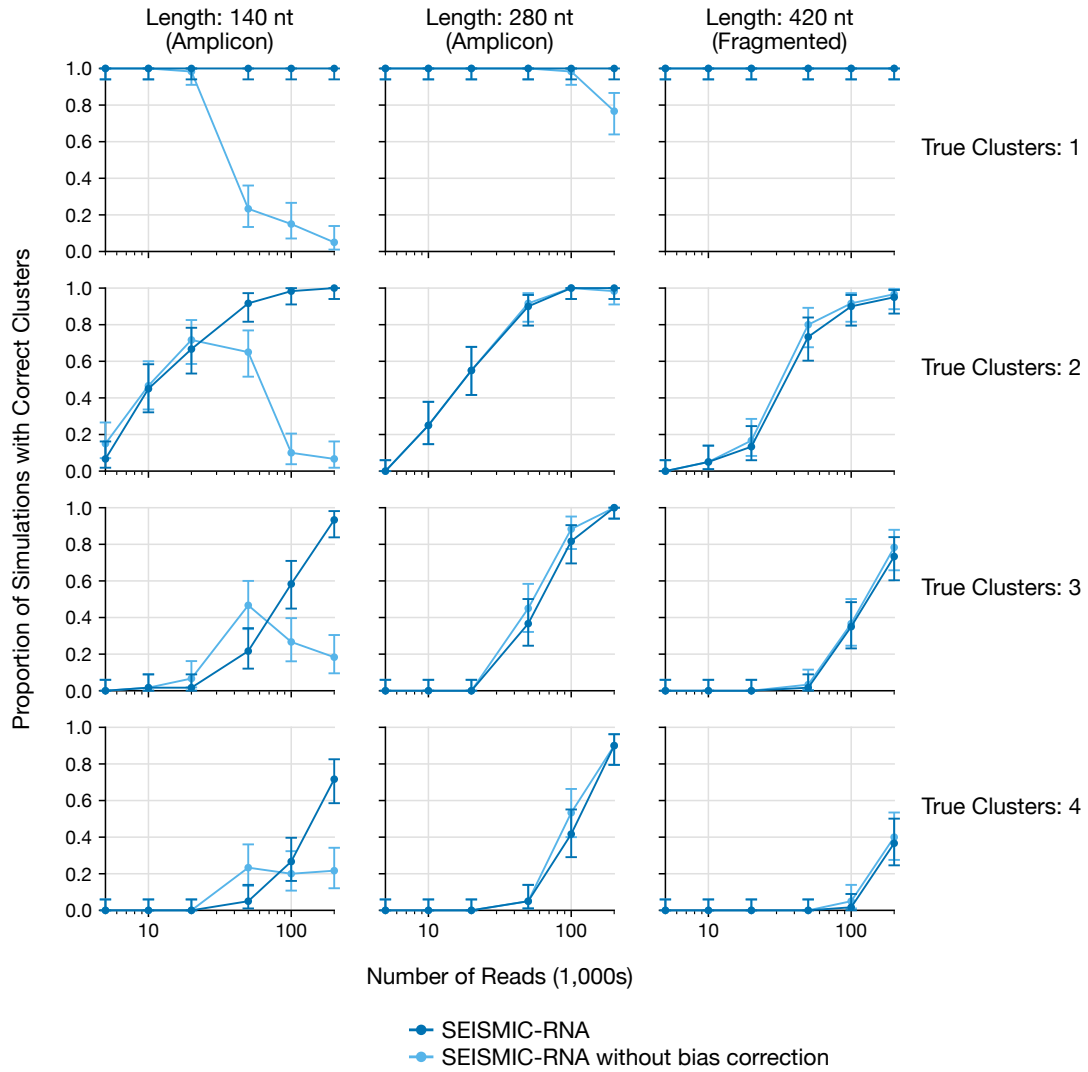
Supplementary Figure 4: **Read coverage per position using 200,000 reads.** For each reference length, the number of informative base calls at each position for all 240 RNAs with 200,000 simulated reads (transparent blue). The black line shows the average.



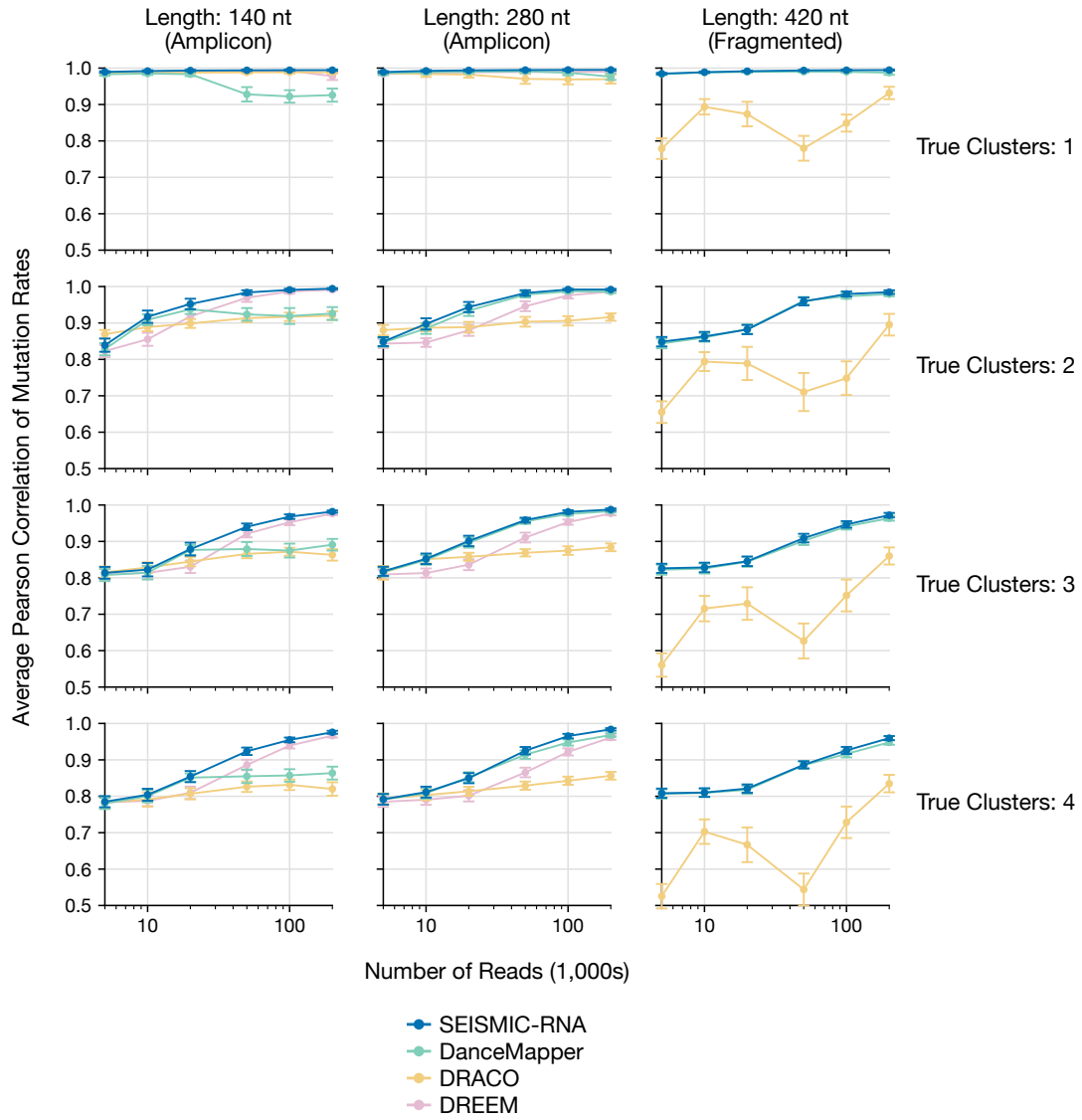
Supplementary Figure 5: **Accuracy of cluster proportions using 200,000 reads.** Average root-mean-square error (RMSE) between the true cluster proportions and those calculated by SEISMIC-RNA, DanceMapper, DRACO, and DREEM for each length of RNA and true number of clusters. Lower RMSEs are more accurate. Error bars show 95% confidence intervals.



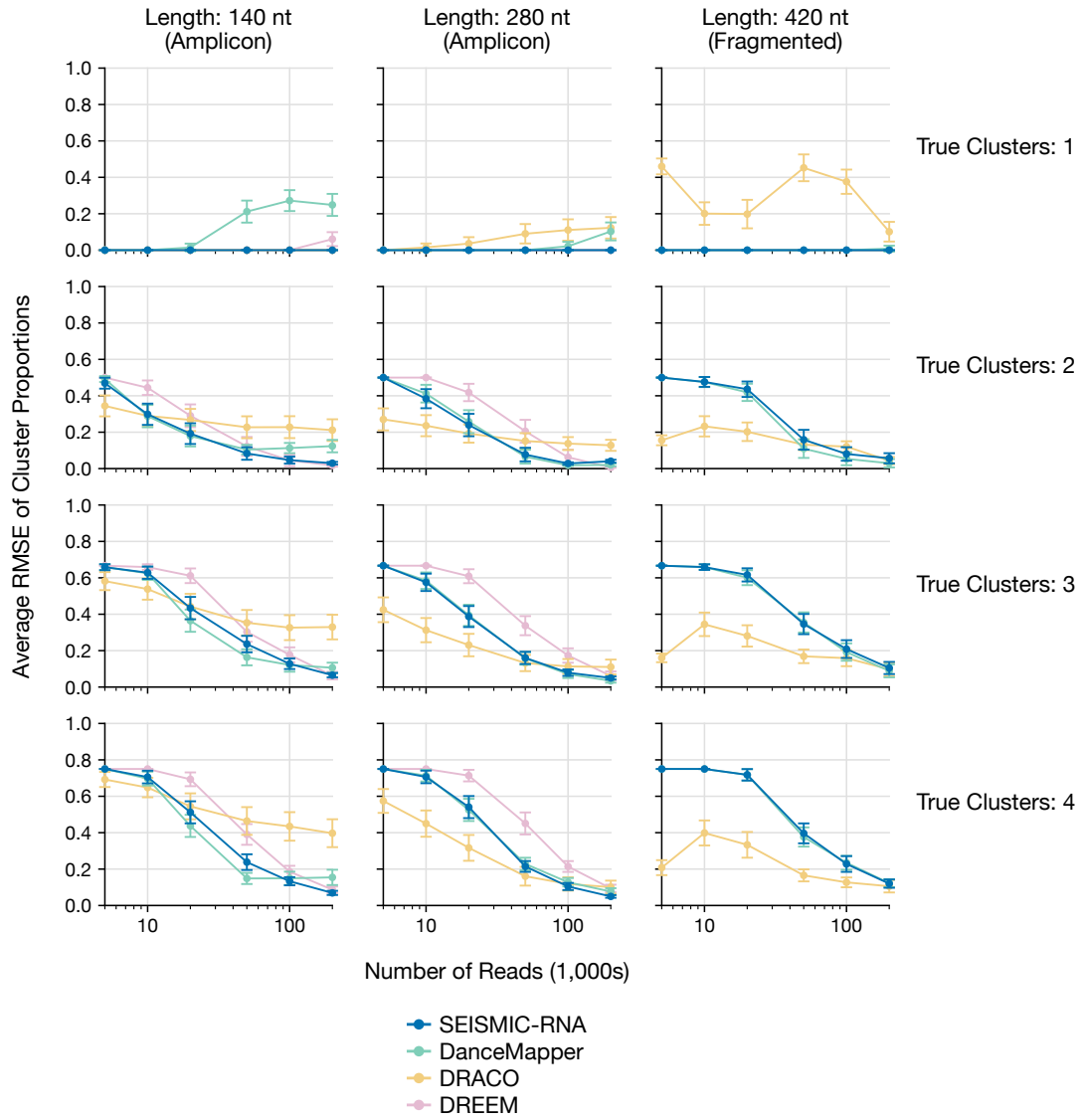
Supplementary Figure 6: **Accuracy of number of clusters versus number of reads for each software.** Proportion of simulated FASTQ files for which each piece of software detected the correct number of clusters as a function of number of reads for each reference length and true number of clusters. Higher proportions are more accurate. Error bars show 95% confidence intervals.



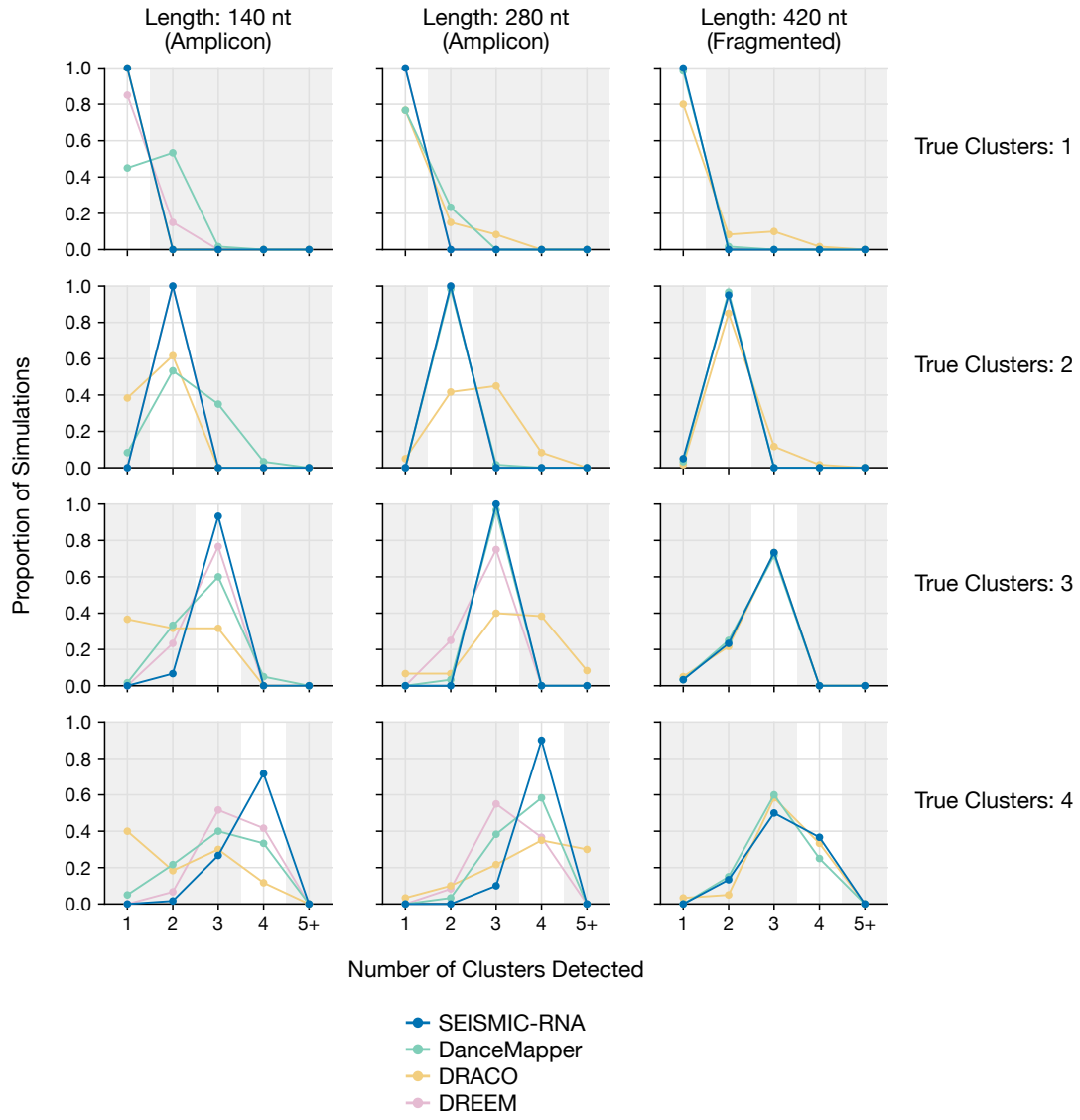
Supplementary Figure 7: **Accuracy of number of clusters versus number of reads without bias correction.** Proportion of simulated FASTQ files for which SEISMIC-RNA with and without bias correction detected the correct number of clusters as a function of number of reads for each reference length and true number of clusters. Higher proportions are more accurate. Error bars show 95% confidence intervals.



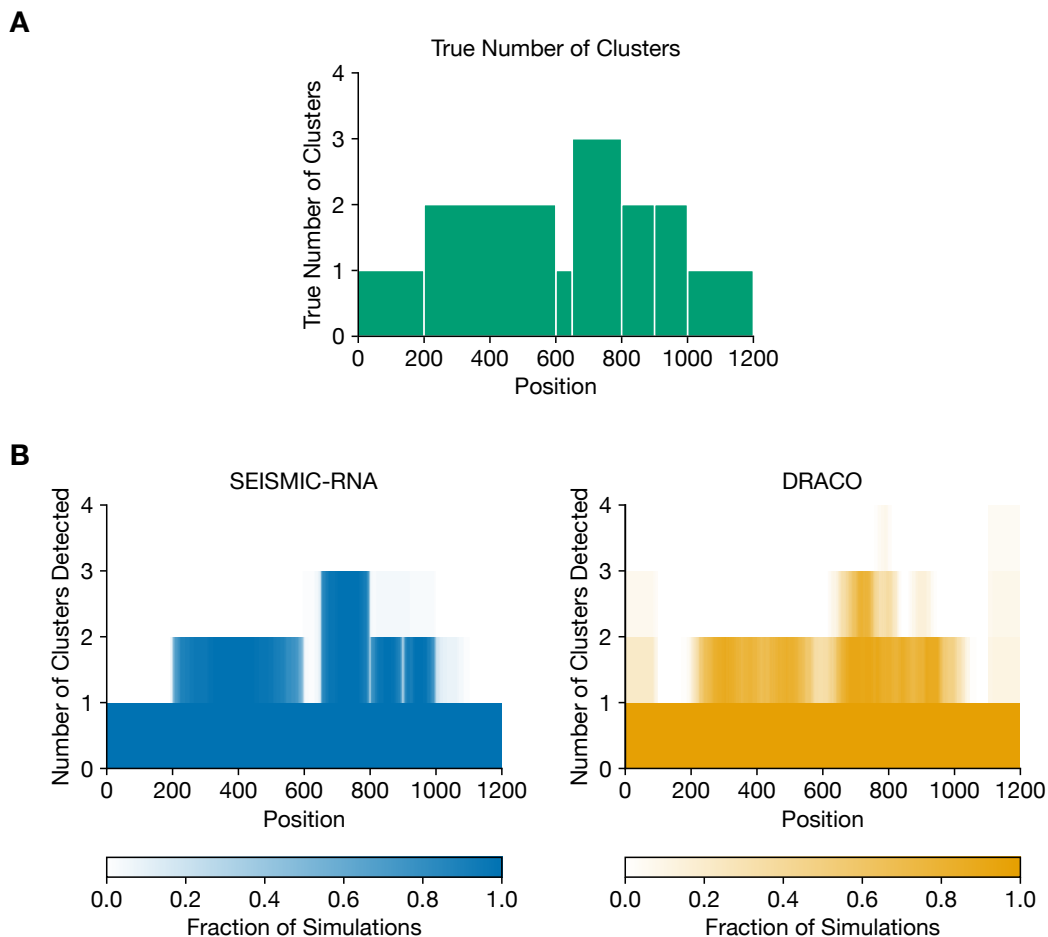
Supplementary Figure 8: **Accuracy of mutation rates versus number of reads for each software.** Average Pearson correlation between the true mutation rates and the mutation rates detected by each piece of software as a function of number of reads for each reference length and true number of clusters. Higher correlations are more accurate. Error bars show 95% confidence intervals.



Supplementary Figure 9: **Accuracy of cluster proportions versus number of reads for each software.** Average root-mean-square error (RMSE) between the true cluster proportions and the cluster proportions detected by each piece of software as a function of number of reads for each reference length and true number of clusters. Lower RMSEs are more accurate. Error bars show 95% confidence intervals.



Supplementary Figure 10: **Distribution of number of clusters detected using 200,000 reads.** Proportion of simulated FASTQ files for which each piece of software detected each number of clusters for each reference length and true number of clusters.



Supplementary Figure 11: **Number of clusters detected at each position in long transcripts.** **(A)** Location, length, and number of clusters formed by each ground truth domain in every 1,200 nt RNA. Domains are surrounded by white borders to help distinguish them visually. **(B)** Fraction of 60 simulated 1,200 nt RNAs for which SEISMIC-RNA and DRACO detected each number of clusters at each position. Cluster numbers are cumulative: e.g. if at position 1, 25% of RNAs yielded 1 cluster, 50% yielded 2, and 25% yielded 3, then the area between 2 and 3 on the y axis (meaning at least 3 clusters) would be 25% dark, between 1 and 2 (meaning at least 2 clusters) would be 75% dark, and between 0 and 1 (meaning at least 1 cluster) would be 100% dark.

Legend					
		SEISMIC-RNA	DanceMapper	DRACO	DREEM
Platform and Installation					
Runs on Linux		■	■	■	■
Runs on macOS (both Intel and ARM)		■	■	■	■
Install with Git or from GitHub		■	■	■	■
Install with Conda or Mamba		■	■	■	■
Install dependencies automatically		■	■	■	■
Inputs					
Accepts raw reads in FASTQ format		■	■	■	■
Accepts raw reads in gzipped FASTQ format		■	■	■	■
Accepts aligned reads in BAM format		■	■	■	■
Handles paired-end reads		■	■	●	■
Handles spliced reads (i.e. introns removed)		●	■	■	■
Data Selection and Filtering					
Separate plus- and minus-strands		■	■	■	■
Trim adapters and low-quality base calls		■	●	■	■
Exclude low-quality base calls from counts		■	■	■	■
Exclude specific types of mutations		■	■	■	■
Exclude positions highly mutated in untreated		■	■	■	■
Exclude arbitrary positions		■	■	■	■
Filter out low-quality positions and reads		■	■	■	■
Filter out reads that align too far from primers		■	■	■	■
Downsample reads to even out coverage		■	■	■	■
Data Merging					
Pool replicates <i>without</i> copying raw data		■	■	■	■
Join regions <i>without</i> counting overlaps twice		■	■	■	■
Join regions <i>after</i> clustering them individually		■	■	■	■
Correction of Artifacts and Biases					
Correct bias from local alignment		■	■	■	■
Handle ambiguous insertions/deletions		■	■	■	■
Correct bias from mutations too close		■	■	■	■
Calculate read bias (jackpotting)		■	■	■	■
Correct errors in reference sequences		■	■	■	■
Clustering					
Cluster short RNAs or short regions of RNAs		■	■	■	■
Cluster long RNAs in their entirety		■	■	■	■
Data Analysis and Normalization					
Calculate correlations between samples		■	■	■	■
Calculate correlations between positions		■	■	■	■
Calculate ROC curves and AUC-ROC		■	■	■	■
Calculate Gini index		■	■	■	■
Calculate Shannon entropy		■	■	■	■
Normalize reactivities using a percentile		■	■	■	■
Normalize reactivities using untreated sample		■	■	■	■
Normalize reactivities per type of nucleotide		■	■	■	■
RNA Structure Modeling					
Predict RNA structures using reactivities		■	■	■	■
Predict pseudoknots		■	■	■	■
Optimize SHAPE slope/intercept parameters		■	■	■	■
Outputs					
Output results as tabular data (e.g. CSV files)		■	■	■	■
Output results as graphs		■	■	■	■
Draw RNA structures		■	■	■	■
Write reports of parameters and results		■	■	■	■
Write log files		■	■	■	■
Simulation					
Simulate raw FASTQ files		■	■	■	■
Simulate preprocessed data		■	■	■	■
Performance and User Friendliness					
Run entire workflow with one command		■	■	■	■
Process entire directories with one command		■	■	■	■
Preprocess samples/references in parallel		■	■	■	■
Cluster samples/references in parallel		■	■	■	■
Build index for each reference automatically		■	■	■	■
Download prebuilt index for each reference		■	■	■	■
Includes Python interface (importable module)		■	■	■	■

Explanation for "Partially"

Only if mate pairs are merged first
Must input reads in BAM format

Trims low quality, but not adapters

Must create a mask file manually

Pools counts, but not reads

Misses complex ambiguities
Handles only PCR amplicons
Graphs, but does not quantify

Parallelizes only references

Supplementary Figure 12: **SEISMIC-RNA** offers unique features compared to other software. (Continued on next page.)

Supplementary Figure 12: (Continued from previous page.) A list of features supported by each piece of software. For DanceMapper [2] and DRACO [3], this chart includes the features of ShapeMapper 2 [4] and RNA Framework [5], respectively, which are required for preprocessing the data before clustering. For each feature that is “Partially” supported, an explanation of what specifically is supported is given in the right column.

Supplementary Methods

Converting alignment maps into matrices of relationships

Encoding relationships between the read and reference

There are eight types of primary relationships between a base call in the read and the corresponding position in the reference sequence. These relationships are encoded as bits in an 8-bit (1-byte) integer:

Primary Relationship	Binary (8-bit)	Decimal
Match	00000001	1
Deletion in the read	00000010	2
5' of an Insertion in the read	00000100	4
3' of an Insertion in the read	00001000	8
Substitution to A	00010000	16
Substitution to C	00100000	32
Substitution to G	01000000	64
Substitution to T	10000000	128

Ambiguous relationships can be represented via the bitwise OR of primary relationships. A relationship code signifies that the relationship could be any of the bits of which it is made. For example, if a base call in the read has a low quality score, then it is assumed that the true base in the read could be any of the four bases. If the base at the corresponding position in the reference is an A, then the base call in the read could be either a match (if it were truly an A) or a substitution to C, G, or T (if it were truly any of those bases). Hence, the relationship code is 11100001: the bitwise OR of match and substitutions to C, G, and T. There are several common ambiguous relationships:

Ambiguous Relationship	Binary (8-bit)	Decimal
Paired-end mates contradict each other	00000000	0
Ambiguously Match or Deletion	00000011	3
Ambiguously Match or 5' of an Insertion	00000101	5
Ambiguously Match or 3' of an Insertion	00001001	9
Read is low-quality or N; reference is T	01110001	113
Read is low-quality or N; reference is G	10110001	177
Read is low-quality or N; reference is C	11010001	209
Read is low-quality or N; reference is A	11100001	225
Reference is N (ambiguous/unknown)	11110001	241
The read does not cover the position	11111111	255

The relationship between a base in the reference and a base in the read is determined as follows:

1. If the reference base is not A, C, G, or T, then the relationship code is 11110001.
2. Otherwise, if the read base is not A, C, G, or T; or if its quality score is below the minimum, then the relationship code is 11110001 minus the bit corresponding to a substitution to the reference base (from the table above).
3. Otherwise, if the read base matches the reference base, then the relationship code is 00000001.
4. Otherwise, the relationship code is a substitution to the base in the read (from the table above).

Storing mutational profiling data in a sparse structure

In chemical probing experiments, mutations occur at low rates. Consequently, a sparse data structure is much more efficient at storing mutational profiling data than a full matrix of reads by positions – especially when the reference is much longer than the reads.

The positions of the ends of the reads are stored as two NumPy arrays [6] – one for the 5' and one for the 3' ends. Each row in the array corresponds to one read in a batch, and each column corresponds to one segment in the read: single-end reads have one segment, paired-end reads have two. Additionally, using `seismic join` causes the arrays to be concatenated horizontally, increasing the number of segments per read.

The relationships between the read and reference are stored as a three-level data structure:

1. The first level is a Python `dict` in which each key is a position in the reference sequence and each value is a Python `dict`.
2. In the second level, for each Python `dict` each key is a relationship code (see “Encoding relationships between the read and reference”) and each value is a NumPy array [6].
3. In the third level, each NumPy array contains the indexes of the reads in the batch that have the relationship code (second level) at the position (first level).

Converting aligned reads into vectors of relationships

Each alignment map is collated and converted into a SAM file using SAMtools [7]. Each read in the SAM file is then parsed to generate a vector of relationships to the reference sequence as follows:

1. Initialize the position in the read to 1 and the position in the reference to the mapping position in the SAM file [8].
2. Initialize the 5’ and 3’ ends of the read to 1 and the read length (in read coordinates).
3. Initialize empty lists of the 5’ and 3’ ends of segments in the read.
4. Initialize an empty map of positions to their relationship codes (only for non-matches, to save effort).
5. Initialize an empty list of insertions and deletions (“indels”).
6. Loop over each operation in the CIGAR string [8] and act based on the type of operation:

Match or Substitution (M/X/=): For the length of the operation:

- (a) Use the base in the reference, the base in the read, and the read quality to encode the relationship between the read and the reference (for details, see “Encoding relationships between the read and reference”).
- (b) Map the current position in the reference to the relationship if it is not a match.
- (c) Increment the positions in both the read and reference.

Deletion (D): For the length of the operation:

- (a) Map the current position in the reference to the deletion relationship (for details, see “Encoding relationships between the read and reference”).

- (b) Add a deletion to the list of indels.
- (c) Increment the position in the reference.

Insertion (I): For the length of the operation:

- (a) Add an insertion to the list of indels. (Do not map the reference position to an insertion yet; that will happen later.)
- (b) Increment the position in the read.

Intron or other Skipped Region (N): This is the end of a contiguous segment:

- (a) Add the current position in the reference to the list of segment 3' ends.
- (b) Advance the reference position by the length of the operation.
- (c) Add the new position in the reference to the list of segment 5' ends (so that a new segment can begin).

Soft Clip (S): The 5' or 3' end must be clipped off:

- (a) Determine the location of the soft clip:
 - 5' end:** Increase the 5' end of the read by the length of the operation (to clip off the first bases).
 - 3' end:** Decrease the 3' end of the read by the length of the operation (to clip off the last bases).
 - Otherwise:** The location is invalid. Raise an error.
- (b) Increase the position in the read by the length of the operation.

Otherwise: The operation is not supported. Raise an error.

7. Add any insertions to the map of positions to relationships. Insertions must be added last because they can overlap other types of relationships. For each insertion:
 - (a) Determine which reference position to mark: the 3' (default) or 5' side of the insertion. Each insertion is located between two positions in the reference; only one is marked to avoid double-counting. If marking the 5' side, use the code 00000100; if the 3' side, use 00001000 (for an explanation, see "Encoding relationships between the read and reference").
 - (b) If the position to be marked already appears in the map because it has another mutation, then add the insertion code on top of the existing relationship code using bitwise OR. For example, if the position 3' of an insertion were already a substitution to T (10000000), then the result would be 10001000 after marking the insertion.

8. Mark ambiguous indels (for details, see “Detecting and labeling ambiguous insertions and deletions”).
9. Determine the positions of the 5’ and 3’ ends of the read’s footprint in the reference. Clip a user-specified number of bases n from both ends of the reference to counteract the bias against mutations near the ends of reads that happens during local alignment (for an explanation, see “Calculate relationships between each read and reference” in the Materials and Methods):
 - (a) Set the 5’ end to the smaller of
 - the read’s mapping position in the SAM file plus n
 - the reference length plus 1
 - (b) Set the 3’ end to the larger of
 - the last position reached in the reference minus n
 - 0
10. Prepend the 5’ end of the read’s footprint in the reference to the the list of segment 5’ ends – it begins the first segment of the read. Append the 3’ end of the read’s footprint in the reference to the list of segment 3’ ends – it ends the last segment of the read.
11. Remove from the relationship map any positions that are no longer contained in any segment after clipping n positions from both ends.

Merging the relationships of paired-end reads

If the reads are paired-end and two mates are properly paired, then after both mates are processed as described in “Converting aligned reads into vectors of relationships”, several more steps are required to merge them:

1. Ensure that one mate (can be either 1 or 2) aligns in the forward orientation and the other aligns in the reverse orientation (if not, raise an error).
2. If (optionally) removing overhangs of mates that dovetail:
 - (a) For each segment of the reverse-aligned read, if its 5’ end extends beyond the forward-aligned read, then trim it so that it is flush with the 5’ end of the forward-aligned read. Remove any position-relationship mappings in the trimmed part.
 - (b) For each segment of the forward-aligned read, if its 3’ end extends beyond the reverse-aligned read, then trim it so that it is flush with the 3’ end of

the reverse-aligned read. Remove any position-relationship mappings in the trimmed part.

3. Concatenate the segment 5' and 3' end positions from both mates, putting the segment(s) of the forward-aligned mates first and those of the reverse-aligned mates last.
4. Initialize a merged map of positions to relationships. For each position that occurs in the position-relationship map of at least one mate:
 - (a) Find the relationship at that position in the forward-aligned mate.
 - If the position is in the forward mate's map (because it is a mutation), then use the relationship code from the map.
 - If the position is not in the map but lies within any segment of the forward mate, then the position is a match (code 00000001).
 - Otherwise, the position is not covered by the mate (code 1111111).
 - (b) Find the relationship at that position in the reverse-aligned mate using the same procedure.
 - (c) Take the bitwise AND of the forward and reverse mates' relationship codes. This limits the possible relationships to those on which the mates agree. It also lets one mate compensate for low quality or lack of coverage in the other mate. For example, if the reference base was an A, mate 1 was low quality (code 11100001), and mate 2 was a high-quality substitution to T (code 10000000), then the bitwise AND would be a substitution to T.
 - (d) If the bitwise AND is not a match (code 00000001), then add it to the merged map of positions to relationships.

Detecting and labeling ambiguous insertions and deletions

When insertions and deletions (collectively “indels”) occur in repetitive regions or next to substitutions or low-quality base calls, it can be impossible to assign them unambiguously to one position. For example, if the reference sequence were AGCCTA and a read were AGCTA, then it would be clear that one C had been deleted, but impossible to determine which. Aligners like Bowtie 2 [9] will arbitrarily choose a location for the indel, but it may not be the true location. To avoid labeling the wrong positions as indels, all indels with ambiguous locations must be identified and labeled.

First, all indels and their locations are listed during the relationship encoding process (for details, see “Converting aligned reads into vectors of relationships”. Each indel keeps track of two positions:

Inline position: The position in the sequence that contains the base. It must be **unique** among indels.

For a deletion, this is the position in the **reference** of the base that has been deleted from the read.

For an insertion, this is the position in the **read** of the base that has been inserted into the read.

Lateral position: The position in the sequence that is missing the base. Because the base is absent, it does not correspond to a position in the sequence, so the position of the existing base on either side must be used instead (hence “lateral” position). By convention the position of the base immediately 3’ of the missing base is used as the lateral position. Adjacent indels will have the **same** lateral position.

For a deletion, this is the position in the **read** of the base immediately 3’ of the deletion.

For an insertion, this is the position in the **reference** of the base immediately 3’ of the insertion.

Indels of the same type (namely, insertion or deletion) that do not have another indel of the other type between them are grouped into “pods”. Indels do not need to occur at consecutive positions to be in the same pod: there just needs to be no indel of the other type between them. For example, suppose a read had deletions at reference positions 20, 45, and 103; and insertions at positions 68 and 72. It would have three pods:

1. Deletions 20 and 45
2. Insertions 68 and 72
3. Deletion 103

The algorithm discovers all possible ways the indels could move without changing the read-reference relationships (hence why the locations are ambiguous) under the constraint that two pods never move past each other. (This constraint is necessary for computational tractability because it allows processing each pod independently; in practice it almost never makes a difference in the result.) It works as follows, in two major phases:

1. In the first phase, all indels are moved as far as possible in the 5' direction. To avoid one pod blocking another's movement, the 5'-most pod must be moved first. For each pod in order from 5' to 3':

- (a) Ensure the indels in the pod are sorted from 5' to 3'.
- (b) Given that the pod contains n indels, initialize a pointer i to indel n (the 3'-most indel). The 3'-most indel must be moved first in order to handle multi-base deletions.
- (c) While $i \geq 1$ (the 5'-most indel):

- i. Compute a set of three trial positions to which to try moving indel i :

Trial inline position: Set it to one position 5' of the indel's current inline position. Because inline positions must be unique, keep moving it one position 5' until it does not match the inline position of any other indel. This prevents the trial inline position from matching the inline position of any adjacent indel.

Trial lateral position: Set it to the position 5' of the indel's current lateral position.

Trial swap position: The position in the sequence that is missing the base with which the indel is trying to swap. Set it to the lateral position on the 5' side of the indel.

- ii. Check whether the trial move is valid, if all of the following are true:

- The trial inline position is within the bounds of the reference sequence (if a deletion) or read sequence (if an insertion).
- The move would not cause the indel to collide with any indel in an adjacent pod, based on the trial lateral position.
- The relationships would be consistent before and after moving the indel:
 - A. Find the base with which the indel would swap during the move (located at the trial swap position).

For a deletion: the base in the read immediately 5' of the site of the deletion.

For an insertion: the base in the reference immediately 5' of the site of the insertion.

- B. Find the base at the current inline position.

For a deletion: the base in the reference that is deleted from the read.

For an insertion: the base that is inserted into the read.

C. Find the base at the trial inline position.

For a deletion: the base in the reference that would be marked as deleted if the deletion moved.

For an insertion: the base in the read that would be marked as inserted if the insertion moved.

D. Using the method in “Encoding relationships between the read and reference”, compute the relationships between the swap base and

- the base at the current inline position
- the base at the trial inline position

E. The relationships are consistent if either of the following is true:

- The relationship codes have at least one bit in common (the bitwise AND is not zero).
- Both relationships are substitutions to any base (not necessarily to the same base).

iii. If the move is valid, then move the indel to the trial position.

For a deletion: The deletion moves from the current inline position to the trial inline position in the reference. Meanwhile, the swap base in the read moves from the trial inline position to the the current inline position.

A. To the current inline position, mark (with bitwise OR) the relationship between it and the swap base, which moves to that position.

B. To the trial inline position, mark (with bitwise OR) the code for the deletion (00000010), which moves to that position.

C. Move the deletion’s inline and lateral positions to the trial positions.

For an insertion, if marked on the 5’ side: The insertion moves from the current inline position to the trial inline position in the read. Meanwhile, the swap base in the reference moves from the trial inline position to the the current inline position.

A. Move the insertion’s inline and lateral positions to the trial positions.

- B. On the 5' side after the move, mark (with bitwise OR) the code for the insertion.
- C. On the 3' side after the move, mark (with bitwise OR) the relationship between the current inline position and the swap base. If that relationship is a match, then mark it only if that position is not immediately 5' of another insertion.

For an insertion, if marked on the 3' side: The insertion moves from the current inline position to the trial inline position in the read. Meanwhile, the swap base in the reference moves from the trial inline position to the the current inline position.

- A. On the 3' side before the move, mark (with bitwise OR) the relationship between the read and reference at that position. (If that relationship is a match, then it will have been marked as 3' of an insertion, not a match, so because the insertion is moving it must now also be marked as a match.)
- B. Move the insertion's inline and lateral positions to the trial positions.
- C. On the 3' side after the move, mark (with bitwise OR) the codes for the insertion and for the relationship between the current inline position and the swap base (if it's not a match).

iv. If any indel moved through another indel, then re-sort the indels in the pod so that they are still in 5' to 3' order.

v. If the move was not valid, then decrement i . This means that this indel has no more valid moves in the 5' direction, so the next indel must be tried.

2. In the second phase, all indels are moved as far as possible in the 3' direction. Unlike in the first phase, also detect all possible permutations of indels as they move. It works the same way as the first phase, with two exceptions:

- Movements and sorting are in the opposite direction: all 5' and 3' labels in the first phase are swapped.
- If an indel is moved successfully, then adjust i to point at that indel again, rather than keeping i unchanged. For example, if i were 1 and the indel in 1st place in the pod wound up in 2nd place after it moved and the indels were re-sorted, then adjust i to 2. This adjustment is necessary to consider all possible

permutations of indels in the second phase, which searches more exhaustively than the first phase.

Correcting bias due to underrepresentation of reads with nearby mutations

Let N reads from K clusters align to a reference sequence of length J . Let the proportion of reads whose 5' and 3' ends align, respectively, to coordinates a and b ($1 \leq a \leq b \leq J$) be η_{ab} (assuming these proportions are equal for all clusters). Let the mutation rate of base j ($1 \leq j \leq J$) in cluster k ($1 \leq k \leq K$) be μ_{jk} . Let the proportion of cluster k in the ensemble be π_k . To express these quantities as probabilities, let C_k be the event that a read comes from cluster k ; let E_{ab} be the event that a read aligns with 5' and 3' coordinates a and b , respectively; let S_j be the event that a read contains position j (i.e. its alignment coordinates a and b satisfy $1 \leq a \leq j \leq b \leq J$); let M_j be the event that a read has a mutation at position j ; and let G_g be the event that a read has no two mutations separated by fewer than g non-mutated bases.

Deriving mutation rates of reads with no two mutations too close

In terms of these events, the total mutation rates (μ_{jk}) are $P(M_j|S_jC_k)$, i.e. the probability that a read would have a mutation at position j given that it contained position j and came from cluster k ; and the observable mutation rates (m_{jk}) are $P(M_j|S_jC_kG_g)$, i.e. the probability that a read would have a mutation at position j given that it contained position j , came from cluster k , and had no two mutations closer than g bases. Using these definitions and Bayes' theorem yields a probabilistic formula for m_{jk} :

$$m_{jk} = P(M_j|S_jC_kG_g) = P(M_j|S_jC_k) \frac{P(G_g|S_jM_jC_k)}{P(G_g|S_jC_k)} = \mu_{jk} \frac{P(G_g|S_jM_jC_k)}{P(G_g|S_jC_k)}$$

The term $P(G_g|S_jC_k)$ is the probability that a read would have no two mutations closer than g bases given that it contained position j and came from cluster k . It can be computed using $P(G_g|E_{ab}C_k)$ (abbreviated d_{abk}): the probability that a read would contain no two mutations closer than g bases given that its 5' and 3' coordinates are a and b , respectively ($1 \leq a \leq b \leq J$), and that it came from cluster k . If position b were mutated (probability μ_{bk}), then the read would contain no two mutations closer than g bases if and only if none of the g bases preceding b (i.e. positions $b - g$ to $b - 1$, inclusive)

were mutated (probability $\prod_{j'=\max(b-g,a)}^{b-1} (1 - \mu_{j'k})$, abbreviated $w_{\max(b-g,a),b-1,k}$) and no two mutations between positions a and $b - (g + 1)$, inclusive, were too close (probability $d_{a,\max(b-(g+1),a),k}$). If position b were not mutated (probability $1 - \mu_{bk}$), then the read would contain no two mutations closer than g bases if and only if no mutations between positions a and $b - 1$, inclusive, were too close (probability $d_{a,\max(b-1,a),k}$). These two possibilities generate a recurrence relation:

$$d_{abk} = \mu_{bk} w_{\max(b-g,a),b-1,k} d_{a,\max(b-(g+1),a),k} + (1 - \mu_{bk}) d_{a,\max(b-1,a),k}$$

The base case is $d_{abk} = 1$ when $a = b$ because such a read would contain one position and thus be guaranteed to have no two mutations too close. Then, $P(G_g|S_j C_k)$ is the average of d_{abk} over every read that contains position j , weighted by the proportions η_{ab} :

$$P(G_g|S_j C_k) = \frac{\sum_{a=1}^j \sum_{b=j}^J \eta_{ab} d_{abk}}{\sum_{a=1}^j \sum_{b=j}^J \eta_{ab}}$$

The term $P(G_g|S_j M_j C_k)$ is the probability that a read would have no two mutations too close given that it contained a mutation at position j and came from cluster k . It can be computed using $P(G_g|M_j E_{ab} C_k)$ (abbreviated f_{abjk}): the probability that a read would contain no two mutations too close given that position j is mutated ($1 \leq a \leq j \leq b \leq J$), that its 5' and 3' coordinates are a and b (respectively), and that it came from cluster k . Because position j is mutated, having no two mutations too close requires that none of the g bases on both sides of position j be mutated. The probability that none of the preceding g positions ($j - g$ to $j - 1$) is mutated is $w_{\max(j-g,a),j-1,k}$, while that of the following g positions ($j + 1$ to $j + g$) is $w_{j+1,\min(j+g,b),k}$. Upstream of the g bases flanking position j (i.e. positions a to $j - (g + 1)$), the probability that no two mutations are too close is $d_{a,\max(j-(g+1),a),k}$; downstream (i.e. positions $j + (g + 1)$ to b), the probability is $d_{\min(j+(g+1),b),b,k}$. Since mutations in these four sections are independent, the probability that the read contains no two mutations too close is the product:

$$f_{abjk} = d_{a,\max(j-(g+1),a),k} w_{\max(j-g,a),j-1,k} w_{j+1,\min(j+g,b),k} d_{\min(j+(g+1),b),b,k}$$

Then, $P(G_g|S_jM_jC_k)$ is the average of f_{abjk} over every read that contains position j , weighted by the proportions η_{ab} .

$$P(G_g|S_jM_jC_k) = \frac{\sum_{a=1}^j \sum_{b=j}^J \eta_{ab} f_{abjk}}{\sum_{a=1}^j \sum_{b=j}^J \eta_{ab}}$$

Combining the above results yields an explicit formula for m_{jk} :

$$m_{jk} = \mu_{jk} \frac{\sum_{a=1}^j \sum_{b=j}^J \eta_{ab} f_{abjk}}{\sum_{a=1}^j \sum_{b=j}^J \eta_{ab} d_{abk}}$$

Deriving end coordinate proportions of reads with no two mutations too close

The total proportions (η_{ab}) of reads aligned to 5' and 3' coordinates a and b , respectively, are $P(E_{ab})$; and the proportions of reads with no two mutations too close that align with coordinates a and b (e_{abk}) are $P(E_{ab}|G_gC_k)$. Using these definitions and Bayes' theorem yields a probabilistic formula for e_{abk} :

$$e_{abk} = P(E_{ab}|G_gC_k) = P(G_g|E_{ab}C_k) \frac{P(E_{ab}|C_k)}{P(G_g|C_k)} = d_{abk} \frac{\eta_{ab}}{P(G_g|C_k)}$$

Note that, while reads are assumed to come from the same distribution of coordinates (η_{ab}) regardless of their cluster k , the observable distribution of coordinates (e_{abk}) varies by cluster because d_{abk} and $P(G_g|C_k)$ depend on k . The term $P(G_g|C_k)$ is the probability that a read would have no two mutations too close given that it came from cluster k . It can be computed as an average of $P(G_g|E_{ab}C_k)$ (i.e. d_{abk}) over all coordinates a and b (such that $1 \leq a \leq b \leq J$), weighted by the proportion of each coordinate, $P(E_{ab})$ (i.e. η_{ab}):

$$P(G_g|C_k) = \frac{\sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk}}{\sum_{a=1}^J \sum_{b=a}^J \eta_{ab}} = \sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk}$$

This expression is already normalized because $\sum_{a=1}^J \sum_{b=a}^J \eta_{ab} = 1$, by definition.

Combining the above results yields an explicit formula for e_{abk} :

$$e_{abk} = \frac{\eta_{ab} d_{abk}}{\sum_{a'=1}^J \sum_{b'=a'}^J \eta_{a'b'} d_{a'b'k}}$$

Deriving cluster proportions of reads with no two mutations too close

The proportion of total reads in cluster k is $\pi_k = P(C_k)$. The proportion among only reads with no two mutations closer than g bases is

$$p_k = P(C_k|G_g) = P(G_g|C_k) \frac{P(C_k)}{P(G_g)} = \pi_k \frac{\sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk}}{P(G_g)}$$

The term $P(G_g)$ is the probability that a read from any cluster would have no two mutations closer than g bases and can be solved for by leveraging that the cluster proportions (p_k) must sum to 1:

$$1 = \sum_{k=1}^K p_k = \sum_{k=1}^K \pi_k \frac{\sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk}}{P(G_g)} = \frac{1}{P(G_g)} \sum_{k=1}^K \pi_k \sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk}$$

$$P(G_g) = \sum_{k=1}^K \pi_k \sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk}$$

The result is an explicit formula for p_k :

$$p_k = \frac{\pi_k \sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk}}{\sum_{k'=1}^K \pi_{k'} \sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk'}}$$

Solving total mutation rates and cluster and coordinate proportions

The observed mutation rates (m_{jk}), end coordinate proportions (e_{abk}), and cluster proportions (p_k) can be calculated as weighted averages over the N reads with no two mutations too close:

$$m_{jk} = \frac{\sum_{i=1}^N z_{ik} x_{ij}}{\sum_{i=1}^N z_{ik} s_{ij}}$$

$$e_{abk} = \frac{\sum_{i=1}^N z_{ik} y_{abi}}{\sum_{i=1}^N z_{ik}}$$

$$p_k = \frac{\sum_{i=1}^N z_{ik}}{N}$$

where s_{ij} is 1 if read i contains position j , otherwise 0; x_{ij} is 1 if read i has a mutation at position j , otherwise 0; y_{abi} is 1 if read i aligns to coordinates a and b , otherwise 0; and z_{ik} is the probability that read i came from cluster k .

The original parameters μ_{jk} , η_{abk} , and π_k can be solved by setting the two formulae each for m_{jk} , e_{abk} , and p_k equal to each other, creating a system of equations:

$$\begin{aligned}\mu_{jk} \frac{\sum_{a=1}^j \sum_{b=j}^J \eta_{ab} f_{abjk}}{\sum_{a=1}^j \sum_{b=j}^J \eta_{ab} d_{abk}} &= m_{jk} = \frac{\sum_{i=1}^N z_{ik} x_{ij}}{\sum_{i=1}^N z_{ik} s_{ij}} \\ \eta_{ab} \frac{d_{abk}}{\sum_{a'=1}^J \sum_{b'=a'}^J \eta_{a'b'} d_{a'b'k}} &= e_{ab} = \frac{\sum_{i=1}^N z_{ik} y_{abi}}{\sum_{i=1}^N z_{ik}} \\ \pi_k \frac{\sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk}}{\sum_{k'=1}^K \pi_{k'} \sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk'}} &= p_k = \frac{\sum_{i=1}^N z_{ik}}{N}\end{aligned}$$

Solving this entire system at once has proven computationally impractical for all but extremely short sequences. A more feasible approach is to first solve for μ_{jk} given an initial guess for η_{ab} , next solve for η_{ab} given the updated μ_{jk} , then solve for π_k given the updated μ_{jk} and η_{ab} , and iterate until all three sets of parameters converge.

Even assuming every η_{ab} is a constant, these equations are still too complex to solve for μ_{jk} analytically because d_{abk} and f_{abjk} also depend on μ_{jk} (as well as on other μ variables). Thus, every μ_{jk} is solved for numerically by rearranging each equation to

$$\mu_{jk} \frac{\sum_{a=1}^j \sum_{b=j}^J \eta_{ab} f_{abjk}}{\sum_{a=1}^j \sum_{b=j}^J \eta_{ab} d_{abk}} - m_{jk} = 0$$

and applying the Newton-Krylov method [10] implemented in SciPy [11]. This method requires an initial guess for μ_{jk} ; if that guess is similar to the eventual solution, the algorithm will run faster. When clustering with the EM algorithm (see “Clustering reads with the Expectation-Maximization (EM) algorithm”), every iteration except the first uses the μ_{jk} from the previous iteration as the initial guess, since consecutive iterations tend to have similar parameters. In all other cases, the biased mutation rates (m_{jk}) are used, as they tend to be similar to μ_{jk} unless the mutation rates are large.

Once every μ_{jk} has been solved for, every η_{ab} can be updated. Because d_{abk} does not depend on η_{ab} (except indirectly through the μ_{jk} parameters, which are now assumed to be constants), each equation can be rearranged to

$$\eta_{ab} = \frac{e_{ab}}{d_{abk}} \sum_{a'=1}^J \sum_{b'=a'}^J \eta_{a'b'} d_{a'b'k}$$

Leveraging that $\sum_{a=1}^J \sum_{b=a}^J \eta_{ab} = 1$, by definition, leads to

$$\sum_{a=1}^J \sum_{b=a}^J \frac{e_{ab}}{d_{abk}} \sum_{a'=1}^J \sum_{b'=a'}^J \eta_{a'b'} d_{a'b'k} = 1$$

$$\sum_{a'=1}^J \sum_{b'=a'}^J \eta_{a'b'} d_{a'b'k} = \frac{1}{\sum_{a=1}^J \sum_{b=a}^J \frac{e_{ab}}{d_{abk}}}$$

and finally a closed-form expression for each η_{ab} given μ_{jk} (and hence d_{abk}) and e_{abk} :

$$\eta_{ab} = \frac{\frac{e_{ab}}{d_{abk}}}{\sum_{a'=1}^J \sum_{b'=a'}^J \frac{e_{a'b'}}{d_{a'b'k}}}$$

This equation should theoretically yield the same value of η_{ab} for every k . In practice, the values will differ due to inexactness in floating-point arithmetic. Thus, the consensus value of η_{ab} is taken to be the average η_{ab} over every k , weighted by π_k :

$$\eta_{ab} = \sum_{k=1}^K \pi_k \frac{\frac{e_{ab}}{d_{abk}}}{\sum_{a'=1}^J \sum_{b'=a'}^J \frac{e_{a'b'}}{d_{a'b'k}}}$$

With updated values of μ_{jk} and η_{ab} , π_k can also be solved. The above equations can be rearranged to

$$\pi_k = p_k \frac{\sum_{k'=1}^K \pi_{k'} \sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk'}}{\sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk}}$$

Given that $\sum_{k=1}^K \pi_k = 1$, by definition:

$$\sum_{k=1}^K p_k \frac{\sum_{k'=1}^K \pi_{k'} \sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk'}}{\sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk}} = 1$$

$$\sum_{k'=1}^K \pi_{k'} \sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk'} = \frac{1}{\sum_{k=1}^K \frac{p_k}{\sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk}}}$$

which leads to a closed-form expression for each π_k given μ_{jk} (and hence d_{abk}), η_{ab} , and p_k :

$$\pi_k = \frac{\frac{p_k}{\sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk}}}{\sum_{k'=1}^K \frac{p_{k'}}{\sum_{a=1}^J \sum_{b=a}^J \eta_{ab} d_{abk'}}}$$

Calculating the probability of observing each read

The joint probability L_{ik} that a random would come from cluster k and have specific 5'/3' ends (E_{ab}) and mutations (M) given that no two mutations are too close can be factored

into three parts using the chain rule for probability:

$$L_{ik} = P(E_{ab}MC_k|G_g) = \frac{P(E_{ab}MC_kG_g)}{P(G_g)} = P(M|E_{ab}C_kG_g)P(E_{ab}|C_kG_g)P(C_k|G_g)$$

The first part – the probability that a read would have the specific mutations x_{ij} given that its 5'/3' end coordinates are a and b (respectively), it comes from cluster k , and no two mutations are too close – is the product over every position j from a to b of the probability of a mutation (μ_{jk}) if read i is mutated at position j ($x_{ij} = 1$), otherwise ($x_{ij} = 0$) the probability of no mutation ($1 - \mu_{jk}$), normalized by the probability that no two mutations would be too close (d_{abk}):

$$P(M|E_{ab}C_kG_g) = \frac{1}{d_{abk}} \prod_{j=a}^b \mu_{jk}^{x_{ij}} (1 - \mu_{jk})^{(1-x_{ij})}$$

The second part, $P(E_{ab}|C_kG_g) = e_{abk}$, can be calculated from the parameters μ_{jk} , η_{ab} , and π_k , as explained in “Deriving end coordinate proportions of reads with no two mutations too close”. Likewise, the third part, $P(C_k|G_g) = p_k$, can also be calculated from the parameters, as explained in “Deriving cluster proportions of reads with no two mutations too close”. Combining all parts yields a formula for L_{ik} in terms of the parameters μ_{jk} , η_{ab} , and π_k and of their derived values d_{abk} , e_{abk} , and p_k :

$$L_{ik} = p_k \frac{e_{abk}}{d_{abk}} \prod_{j=a}^b \mu_{jk}^{x_{ij}} (1 - \mu_{jk})^{(1-x_{ij})}$$

Then, the marginal probability of observing read i from any cluster can be expressed as the sum of the joint probability L_{ik} over all clusters:

$$L_i = P(E_{ab}M|G_g) = \sum_{k=1}^K P(E_{ab}MC_k|G_g) = \sum_{k=1}^K L_{ik}$$

Accelerating the bias correction algorithm by processing in chunks

The bias correction algorithm as described above scales with sequence length J as roughly $O(J^2)$ in memory and $O(J^3)$ in time, becoming prohibitive for long sequences. However, an approximate solution can be derived in roughly $O(J)$ time. The approximation works on the basis that if there exists a stretch of at least g consecutive positions with very low mutation rates, then the positions before this stretch will mutate essentially independently

of the positions after it. Thus, a long sequence can be split into multiple chunks at every stretch of g or more consecutive bases with very low mutation rates. Although the runtime for each chunk still scales as $O(J_{chunk}^3)$ with its length J_{chunk} , under the assumption that average chunk length does not depend on total sequence length, the total runtime for the sequence scales linearly with the number of chunks, which scales as linearly with the sequence length J , leading to overall scaling of $O(J)$.

To split the sequence, the first position of and the position immediately after each consecutive stretch of g or more positions below a user-specified threshold μ_t are identified. Then, the observed mutation rates (m) are split at those positions using the `split` function from NumPy [6], producing one chunk for each interval of g or more low mutation rates and one chunk for each interval of higher mutation rates between them. For example, if the pattern of above (A) and below (B) m_t were [A, B, A, B, B, B, B, A, A, B] and g were 4, then the (0-indexed) split positions would be [3, 7] and the sequence would be split into 3 parts: [A, B, A], [B, B, B, B], and [A, A, B].

The end coordinate probabilities (η) must also be split into chunks at the same positions as the mutation rates. For each chunk spanning positions a to b , the probabilities (η_{chunk}) must be adjusted to include reads that extend beyond the ends of the chunk. That is, the first row of the matrix η_{chunk} must represent the probability that a read's 5' end is at *or before* the 5' end of the chunk. Thus, the first row of η_{chunk} is calculated by summing rows 1 to a of η :

$$\eta_{chunk_{1j}} = \sum_{j'=1}^a \eta_{j',(j+a-1)}$$

where a and b are the 5' and 3' ends of the chunk, respectively, and j is between 1 and $b - a$ (inclusive).

Likewise, the last column ($b - a + 1$) is calculated by summing columns b to J of η :

$$\eta_{chunk_{j,(b-a+1)}} = \sum_{j''=b}^J \eta_{(j+a-1),j''}$$

where a and b are the 5' and 3' ends of the chunk, respectively, and j is between 2 and $b - a + 1$ (inclusive).

The last element of the first row of η_{chunk} represents the probability that a read spans the entire chunk:

$$\eta_{chunk_{1,(b-a+1)}} = \sum_{j'=1}^a \sum_{j''=b}^J \eta_{j'j''}$$

All other elements of η_{chunk} (not in the first row or last column) are copied without adjustment from the original matrix η .

The bias correction algorithm described above is performed on each chunk; and then the mutation rates from each chunk are reassembled. Bias correction with and without chunking yield very similar mutation rates for the default threshold of $m_t = 0.001$, but chunking can run orders of magnitude faster for long sequences. Additionally, because DMS-MaPseq data generally exclude G and U bases (the mutation rate is set to 0), and stretches of consecutive Gs/Us are common naturally, most RNA sequences can be divided into a large number of small chunks, providing a large speed-up.

Calculating the jackpotting quotient

The jackpotting quotient (JQ) measures how much the read counts deviate from their expected values relative to how much they would be expected to deviate if there were no jackpotting. Thus, calculating the jackpotting quotient involves first calculating the deviation from expected values (called the “jackpotting score”) for the real dataset. But even without jackpotting, if all positions mutated independently (except those separated by fewer than g bases), read counts would deviate somewhat from the expected values due to the stochasticity of simulation and the finite number of reads. To determine the expected deviation, several null models are produced by simulating reads with independent mutations, calculating the jackpotting scores for all null models, and averaging them. Finally, the jackpotting quotient (JQ) is defined as the quotient of e raised to the real and null jackpotting scores (JS):

$$JQ = \frac{\exp(JS_{real})}{\exp(JS_{null})} = \exp(JS_{real} - JS_{null})$$

Calculating the jackpotting score

The jackpotting score (JS) measures how much the observed read counts deviate from expectations. Because the read counts under the null model (i.e. with independent mutations, except for those separated by less than g bases) follow a multinomial distribution, JS is adapted from the G-test statistic (G^2) [12]

$$G^2 = 2 \sum_{i=1}^N O_i \log \frac{O_i}{E_i}$$

by dropping the coefficient of 2 and dividing by the number of reads N :

$$JS = \frac{1}{N} \sum_{i=1}^N O_i \log \frac{O_i}{E_i}$$

where O_i and E_i are the observed and expected counts of read i , respectively. Thus, the jackpotting score equals the average log-fold overrepresentation of a read.

Calculating the expected count of each read

For read i , the expected count E_i equals the marginal probability of observing read i , which is L_i (for details, see “Calculating the probability of observing each read”) times the total number of reads N :

$$E_i = NL_i$$

Rounding stochastically to integers

Simulating integer numbers of reads or mutations with real number expected values, while keeping the number simulated as close as possible to the original, requires rounding the expected values to the integer above or below. Rounding is performed stochastically, such that the probability of rounding up equals the fractional part of the number (e.g. 5.4 would be rounded up 40% of the time and down 60% of the time).

When there are multiple expected values that represent proportions, the sum after rounding to integers must equal the sum before rounding. (If the sum before rounding is not an integer, then the sum after rounding should be rounded up or down proportional to the fractional part, as described above.) To round all the values while preserving their sum, each is represented as a line segment whose length equals the value. The line segments are shuffled randomly (so that they get rounded up/down independently) and laid out end-to-end on the real number line, with the first segment starting at a random number drawn from the standard uniform distribution (to eliminate bias against the first shuffled value). Then for each line segment, the number of integers on the real number line that it overlaps is counted, which rounds the length of the line segment up (with probability equal to the fractional part of the line segment’s width) or down. Moreover, since the line segments are arranged end-to-end, the total number of integers they overlap must be either the integer immediately less than or greater than the sum of their lengths, which

preserves the sum after rounding. Finally, the rounded values are un-shuffled into their original order.

Simulating a null jackpotting score

The purpose of the null model is to determine the expected jackpotting score when there is no jackpotting. (Note that it is not 0 because reads randomly generated without jackpotting will not exactly match the expected counts – just as a statistic of any finite random sample will not exactly match the parameter of the population.) It works by simulating the same number of reads (N) from the same number of clusters (K) with the same 5' and 3' end coordinates (and hence read coverage), number of mutations per position, and number of reads in each cluster (rounded up or down to the nearest integer) as in the original data. First, the parameters μ_{jk} , η_{ab} , and π_k are used to calculate the biased mutation rates (m_{jk}), 5' and 3' end proportions (e_{abk}), and cluster proportions (p_k) among reads with no mutations separated by fewer than g bases, as described in “Correcting bias due to underrepresentation of reads with nearby mutations”.

To simulate reads with the same 5' and 3' ends as in the real reads, each pair of 5' and 3' ends is assigned probabilistically to a cluster. To do so, the probability that a read with 5' and 3' ends a and b , respectively, came from cluster k is calculated:

$$P(C_k|G_g E_{ab}) = P(E_{ab}|G_g C_k) \frac{P(C_k|G_g)}{P(E_{ab}|G_g)} = e_{abk} \frac{p_k}{\sum_{k'=1}^K e_{abk'} p_{k'}}$$

Then the number of reads in each cluster k is calculated as $N_k = N p_k$ and rounded to an integer stochastically while preserving $\sum_{k=1}^K N_k = N$ (see “Rounding stochastically to integers”). In a randomized order, each pair of 5' and 3' ends is assigned to a cluster:

1. Calculate the probability of assigning pair i of 5' and 3' ends a and b to each cluster k given the number of reads N_k remaining to be assigned: $\frac{N_k P(C_k|G_g E_{ab})}{\sum_{k'=1}^K N_{k'} P(C_{k'}|G_g E_{ab})}$
2. Use those probabilities to select a cluster randomly.
3. Decrement N_k for the cluster k that was selected.
4. Increment i (move to the next pair of 5' and 3' ends) and repeat from step 1.

After each pair of 5' and 3' ends has been assigned to a cluster, reads with those end coordinates are simulated for each cluster. For each cluster k , the read coverage N_{jk} at each position j is calculated from the 5' and 3' coordinates (a_i and b_i , respectively) of

every read i that was assigned to the cluster:

$$N_{jk} = \sum_{i=1}^N a_i \leq j \leq b_i$$

The number of mutations M_{jk} at each position j is calculated as $M_{jk} = m_{jk}N_{jk}$ and stochastically rounded up or down (for details, see “Rounding stochastically to integers”). Then, the positions are sorted from fewest to most non-mutated bases (i.e. $N_{jk} - M_{jk}$) so that positions that are constrained by having few options for non-mutated bases are filled early. The reads that are mutated at each position j are selected as follows:

1. To ensure that in the simulated reads, all mutations are separated at least g bases, all reads with a mutation within g positions of j are listed.
2. The reads that are eligible to be mutated at position j are determined by finding those that cover position j (i.e. $a_i \leq j \leq b_i$) and excluding those that have another mutation within g positions of j .
3. Of those reads, M_{jk} are chosen randomly to be mutated.
4. Finally, j is advanced to the next position, and the process is repeated from step 1.

Finally, the unique simulated reads are counted (the null observed counts, O_i), and the null expected counts (E_i) are calculated as described in “Calculating the expected count of each read”. The jackpotting score of the null model is calculated using the null O_i and E_i as described in “Calculating the jackpotting score”.

Bootstrapping a distribution for the jackpotting quotient

Because the null jackpotting score is stochastic, it will vary among simulations. To ensure that the null jackpotting score is reproducible, the simulation is repeated until it converges, as follows:

1. Initialize an empty list of null jackpotting scores.
2. Simulate a null jackpotting score as described in “Simulating a null jackpotting score” and append it to the list.
3. If length of the list (n) is at least 2, then calculate a user-specified confidence interval (default 95%) for the mean of the list using Student’s t distribution with $n - 1$ degrees of freedom.
4. Use the confidence interval of jackpotting scores to calculate the confidence interval of jackpotting quotients as described in “Calculating the jackpotting quotient”.

5. If the confidence interval of jackpotting quotients contains the user-specified maximum allowable jackpotting quotient, then it is still ambiguous whether or not the reads are too jackpotted: repeat from step 2. Otherwise, return a consensus jackpotting quotient based on the median null jackpotting score.

Calculating the arcsine distance between two sets of mutation rates

The difference between two mutation rates (μ_1 and μ_2) is calculated using the arcsine distance (ARCD):

$$ARCD(\mu_1, \mu_2) = \frac{|\arcsin(2\mu_1 - 1) - \arcsin(2\mu_2 - 1)|}{\pi}$$

The difference between two sets of mutation rates (μ_{jk_1} and μ_{jk_2}) from position $j = a$ to $j = b$ is calculated using the mean arcsine distance (MARCD):

$$MARCD = \frac{1}{b - a + 1} \sum_{j=a}^b ARCD(\mu_{jk_1}, \mu_{jk_2})$$

Arcsine distance is a middle ground between absolute difference, $|\mu_1 - \mu_2|$, which is robust to noise in small but not large mutation rates; and absolute log-fold-change, $|\log(\frac{\mu_1}{\mu_2})|$, which is robust to noise in large but not small mutation rates (and undefined if either mutation rate is 0). Arcsine distance is always in $[0, 1]$, is defined for any $\mu_1, \mu_2 \in [0, 1]^2$, and – for the same absolute difference – is larger when the mutation rates are low (e.g. 0.01 vs. 0.02) than when they are high (e.g. 0.21 vs. 0.22).

Clustering reads with the Expectation-Maximization (EM) algorithm

Let N reads from K clusters align to a reference sequence of length J . Let the proportion of reads whose 5' and 3' ends align, respectively, to coordinates a and b ($1 \leq a \leq b \leq J$) be η_{ab} (assuming these proportions are equal for all clusters). Let the mutation rate of base j ($1 \leq j \leq J$) in cluster k ($1 \leq k \leq K$) be μ_{jk} . Let the proportion of cluster k in the ensemble be π_k .

Maximization step

The maximization step updates the parameters (μ_{jk} , η_{ab} , and π_k) using the current cluster memberships (z_{ik}). The observed estimates of the parameters m_{jk} , e_{ab} , and p_k are first computed; then, the underlying parameters μ_{jk} , η_{ab} , and π_k are solved for as described in “Solving total mutation rates and cluster and coordinate proportions”.

Expectation step

The expectation step updates the cluster memberships (z_{ik}) and the likelihood function (L) using the current parameters (μ_{jk} , η_{ab} , and π_k). Both can be expressed in terms of the joint (L_{ik}) and marginal (L_i) probabilities of observing read i (for details, see “Calculating the probability of observing each read”).

Cluster membership (z_{ik}) is defined as the probability that read i came from cluster k given its 5’/3’ end coordinates (E_{ab}) and mutations (M) and given that no two mutations are too close (G_g):

$$z_{ik} = P(C_k | E_{ab} M G_g) = \frac{P(E_{ab} M C_k G_g)}{P(E_{ab} M G_g)} = \frac{P(E_{ab} M C_k | G_g)}{P(E_{ab} M | G_g)} = \frac{L_{ik}}{L_i}$$

The likelihood of the model (L) is the product of the marginal probabilities (L_i) of observing all the reads:

$$L(\mu, \eta, \pi) = \prod_{i=1}^N L_i = \prod_{i=1}^N \sum_{k=1}^K p_k \frac{e_{abk}}{d_{abk}} \prod_{j=a}^b \mu_{jk}^{x_{ij}} (1 - \mu_{jk})^{(1-x_{ij})}$$

The EM algorithm

Each time the EM algorithm runs, it does the following:

1. Generate K concentration parameters by pulling K random samples from a standard uniform distribution.
2. Initialize the cluster memberships z_{ik} by pulling i random samples from a Dirichlet distribution with the K concentration parameters.
3. Initialize the iteration count to 1.
4. Repeat until convergence or the iteration count exceeds the user-specified maximum limit:
 - (a) Run the Maximization step to update μ_{jk} , η_{ab} , and π_k .

- (b) Run the Expectation step to update z_{ik} and the likelihood L .
 - (c) Break the loop if the solution has converged, which happens if both are true:
 - The current minus the previous $\log L$ is less than a user-specified small positive threshold.
 - The iteration count is at least the user-specified minimum limit.
 - (d) Increment the iteration count.
5. Calculate the Bayesian information criterion [13] (BIC): $BIC = P \log N - 2 \log L$, where P (the number of parameters) is calculated by summing the number of mutation rates in all clusters (K times the number of non-masked positions from 1 to J), the degrees of freedom of the cluster proportions ($K - 1$), and the degrees of freedom of the end coordinate proportions (one less than the number of non-zero elements in the matrix η_{ab} , where $1 \leq a \leq b \leq J$).
6. Check if the result passed all filters:

All clusters are distinct: The maximum Pearson correlation between any two clusters is less than a user-specified maximum OR the minimum mean arcsine distance (see “Calculating the arcsine distance between two sets of mutation rates”) between any two clusters is greater than a user-specified minimum.

No cluster appears to be an artifact: The maximum arcsine distance (see “Calculating the arcsine distance between two sets of mutation rates”) among all positions between the ensemble average and any cluster is less than a user-specified maximum OR the maximum Gini coefficient [14] of any cluster is less than a user-specified maximum.

The run is not jackpotted: The jackpotting quotient (see “Calculating the jackpotting quotient”) is less than a user-specified maximum.

Determining the optimal number of clusters using EM

The EM algorithm is first run once using $K = 1$ to determine the BIC of the ensemble average. Then, the following algorithm finds the optimal K :

1. Increase K by 1.
2. Using the new K , run EM a user-specified minimum number of times (default 6), initializing each run’s cluster memberships (z_{ik}) randomly.
3. Sort all EM runs from highest (best) to lowest (worst) likelihood (L).

4. Check if all of the following conditions are true (to verify that the EM results are reproducible):
 - At least two EM runs passed all filters described in “The EM algorithm”.
 - The maximum Pearson correlation between the best run and any other that passed filters is at least a user-specified minimum.
 - The minimum mean arcsine distance (see “Calculating the arcsine distance between two sets of mutation rates”) between the best run and any other that passed filters is at most a user-specified maximum.
 - The difference in log-likelihood between the best and second-best runs that passed filters is no more than a user-specified maximum.
5. If any conditions were false, then run EM again repeatedly up to a user-specified maximum number of times (default 30). After each additional EM run, repeat steps 3 and 4.
6. Once all conditions have passed or EM has been run the maximum number of times, determine whether the current K is the best encountered so far:
 - (a) For every K that has already been tried, check if all the conditions in step 4 pass. For the current K only, do not require EM runs to pass the jackpotting filter: if the current K is too low, then multiple clusters would still be mixed together, which resembles jackpotting, so failing the jackpotting filter should not preclude trying a larger K .
 - (b) For every K that passed all the conditions, find the best (smallest) BIC among all EM runs that passed filters.
 - (c) Consider the current K to be the best so far if its best run has a smallest BIC than all other values of K .
7. If the current K is the best so far or the user has requested trying all values of K , and the user has not specified a maximum K or the current K is less than the user-specified maximum, then return to step 1 and repeat. Otherwise, stop and use the K that gave the smallest BIC as the optimal number of clusters.

Finding domains that form multiple structures in long transcripts

Transcripts that are much longer than the reads can be clustered by first finding domains in which many pairs of positions are significantly correlated. By definition, positions more

than g bases apart within a cluster do not correlate, so a high density of correlated pairs of positions within a region suggests it forms multiple clusters.

Dividing the reference sequence into tiles

The time and memory required to find correlated pairs of positions scales as $O(J^2)$ where J is the length of the transcript (or a region thereof). Thus, very long transcripts could require prohibitive amounts of computational resources. To speed up the calculation, the transcript is divided into short, overlapping tiles, and correlations are calculated within each tile. By default, the tile length (J_T) is 2 times the median read length, and the minimum overlap fraction (O_T) for adjacent tiles is 0.5. The defaults ensure that nearly all pairs of positions that are both covered by a read – and hence could correlate – appear together in at least one tile, while also keeping the tile size reasonably small – less than 600 nt for 150x150 nt paired-end reads.

The required number of tiles (N_T) is calculated as

$$N_T = 1 + \lceil \frac{J - J_T}{\lfloor J_T(1 - O_T) \rfloor} \rceil$$

where $1 \leq J_T \leq J$ and $0 \leq O_T < 1$; $\lfloor J_T(1 - O_T) \rfloor$ is the maximum permitted offset for adjacent tiles; and $J - J_T$ is the number of positions remaining after the first tile. The 5' ends of the N_T tiles are then spaced evenly from the first position in the transcript/region (J_1) to one tile length before the last position ($J_1 + J - J_T$) using the `linspace` function of NumPy [6] rounded to the nearest integer. Then each 3' end is calculated as the corresponding 5' end plus $J_T - 1$. Finally, the 5' and 3' ends of the tiles are automatically fed into the `mask` step (see “Select and filter data” in the Results and Discussion) to generate the data for each tile.

Finding correlated pairs of positions

After the `mask` step has completed, the data for each tile are processed to calculate confusion matrices. For each pair of positions a and b in the tile for which $b - a > g$ (to ignore pairs of positions separated by fewer than g positions, which tend to be anti-correlated), four quantities are calculated:

N Number of reads that cover both positions a and b .

A Number of reads that have a mutation at position a and cover position b .

B Number of reads that cover position a and have a mutation at position b .

X Number of reads that have a mutation at both positions a and b .

The P -value of each correlation between a pair of positions is calculated using a two-sided Fisher exact test implemented using the hypergeometric distribution in SciPy [11]. The distribution is defined as `dist = hypergeom(N, B, A)`. The lower tail is calculated as `dist.cdf(X)`, equivalent to

$$P_{lo} = \sum_{x=0}^X \frac{\binom{B}{x} \binom{N-B}{A-x}}{\binom{N}{A}}$$

The upper tail is calculated as `dist.sf(X - 1)`, equivalent to

$$P_{up} = 1 - \sum_{x=0}^{X-1} \frac{\binom{B}{x} \binom{N-B}{A-x}}{\binom{N}{A}} = \sum_{x=X}^A \frac{\binom{B}{x} \binom{N-B}{A-x}}{\binom{N}{A}}$$

The two-tailed P -value is then estimated as the 2 times the smaller of the two tails:

$$P = 2 \min(P_{lo}, P_{up})$$

To account for multiple comparisons, significant P -values are found using the Benjamini-Hochberg method [15] with a user-specified false discovery rate α (default 0.05).

Assembling domains from correlated pairs of positions

The pairs of positions with significant correlations are combined across all tiles. Next, domains of the transcript that are enriched for correlated pairs must be identified while tolerating a proportion of false positive pairs (approximately α).

First, all pairs are aggregated into an initial set of domains based on overlap: if two pairs span at least one position in common, then they are placed into the same domain. This process is performed using the following algorithm (in pseudocode):

```
function aggregate_pairs(pairs):
    pairs = sorted(pairs) # pairs must be sorted
    if length(pairs) == 0:
        return {}
    domains = {pairs[0]} # initial domain is first pair
    for (start, end) in pairs[1:]:
        last_domain = domains[length(domains) - 1]
```

```

if start <= last_domain[1]:
    # This pair overlaps the last domain.
    if end > last_domain[1]:
        # This pair extends the last domain.
        last_domain[1] = end
    else:
        # This pair does not overlap the last domain.
        # Create a new domain from the current pair.
        concatenate(domains, {start, end})
return domains

```

The initial aggregation procedure is sensitive to false positives, which can cause distinct domains to be aggregated. To ensure that the domains correspond to dense regions of correlated pairs, the domains are then split if at any position in the domain the total number of pairs that span the position is less than a threshold. To set the threshold, it is assumed that any pair of positions is equally likely to be a false positive, so false positives are distributed uniformly throughout the domain. From this assumption, a formula for the expected number of false positive pairs spanning a position j in a domain (F_j) follows:

$$F_j = \alpha N p_j$$

where α is the user-specified false discovery rate, N is the total number of correlated pairs in the domain, and p_j is the proportion of all possible pairs (a, b) (where $b - a > g$) that span position j :

$$p_j = \frac{\sum_{a=J_1}^j \max(0, (J_D + 1 - \max(j, (a + g + 1))))}{\sum_{a=J_1}^{J_D} \max(0, (J_D + 1 - (a + g + 1)))}$$

where J_1 and J_D are the first and last positions in the domain, respectively, and g is the minimum gap between mutations. The domains are then split as follows:

1. Initialize a list of existing domains to those formed by aggregating all correlated pairs (as described above).
2. Initialize an empty list of finished domains.
3. Initialize the iteration count to 0.
4. Initialize an empty list of new domains for this iteration.
5. For each existing domain spanning positions J_1 to J_D :

- (a) If the domain is already in the list of finished domains, then add it to the list of new domains and continue to the next domain.
 - (b) List all correlated pairs of positions (a, b) contained entirely within the domain (i.e. $J_1 \leq a < b \leq J_D$).
 - (c) If there are fewer than a user-specified minimum number of pairs, then add this domain to the list of finished domains and continue to the next domain.
 - (d) For each position j in the domain, calculate the number S_j of pairs (a, b) that span the position (i.e. $a \leq j \leq b$) and the expected number F_j of false positive pairs that span the position (as described above).
 - (e) If $S_j > F_j$ for every position j in this domain – that is, all positions are expected to be spanned by some true positive pairs – then add this domain to the lists of new domains and finished domains.
 - (f) Otherwise, if $S_j > F_j$ for at least one position j in the domain, then locate the position j_{min} at which $S_j - F_j$ is most negative (where false positive pairs are most in excess). Split the domain at j_{min} into two domains – $(J_1, (j_{min} - 1))$ and $((j_{min} + 1), J_D)$ – eliminating j_{min} itself. Add both to the list of new domains; add the original domain to the list of finished domains.
 - (g) Otherwise, if $S_j > F_j$ for no positions – that is, the entire domain is expected to be false positives – then add this domain to the list of finished domains but not the list of new domains.
6. If the list of new domains is identical to the list of existing domains, then the algorithm has converged. Return the list of existing domains.
 7. Set the list of existing domains equal to the list of new domains.
 8. Increment the iteration count.
 9. If the iteration count exceeds a user-specified threshold, then return the existing domains.
 10. Otherwise, go to step 4 and repeat.

After the domains are constructed, users can optionally include the gaps between domains as follows:

Omit: Omit the gaps between domains; cluster only the domains.

Insert: Insert an extra domain into the gap between each existing domain; cluster all regions of the transcript, even those not expected to form multiple clusters.

Expand: Expand each domain to include part of the gap on each side; each gap is divided evenly between the domains before and after it, with the middle position of an odd-sized gap going to the domain after the gap.

Clustering domains of correlated pairs of positions

A CSV file listing the correlated pairs for each tile and a graph depicting all correlated pairs and domains are generated. The original tiles from the `mask` step are erased, and the new domains are fed into the `mask` step as regions, to generate a dataset for each domain. Finally, each domain is processed with the `cluster` step to solve structure ensembles across the entire transcript.

Joining regions that have been clustered

Solving the optimal matching of clusters from multiple regions

If R regions were clustered with `seismic cluster` and all produced the same number of clusters K , they can be joined with `seismic join`. For each k from 1 to K , a “joined cluster” will be assembled, comprising one cluster from each region r . Because the number assigned to each cluster is arbitrary, the algorithm must determine how to assign each cluster from each region to exactly one joined cluster.

For every pair of regions r_1 and r_2 , the positions in the overlap ($J_{r_1 r_2}$) between the regions are determined. For every combination of clusters k_1 from region r_1 and k_2 from region r_2 , the sum of the arcsine distances (see “Calculating the arcsine distance between two sets of mutation rates”) between the mutation rates (μ) of the overlapping positions *and* of the proportion (π) of each cluster is calculated as the “cost” $C_{r_1 r_2 k_1 k_2}$ of matching up the clusters:

$$C_{r_1 r_2 k_1 k_2} = ARCD(\pi_{k_1}, \pi_{k_2}) + \sum_{j \in J_{r_1 r_2}} ARCD(\mu_{j k_1}, \mu_{j k_2})$$

Including the proportion of each cluster ensures that C_{abxy} is defined for every pair of regions with no overlapping positions, since every cluster has a proportion. Pairs of regions with less (or no) overlap influence the outcome of joining less than pairs of regions with more overlap because they have fewer arcsine distances to sum.

After every cost $C_{r_1 r_2 k_1 k_2}$ is calculated, a hypergraph G is built in which every node N_{rk} represents cluster k in region r . Every hyperedge $E_{k_1 k_2 \dots k_R}$ connects R nodes – one node from each of the R regions – and represents the scenario in which the corresponding clusters from the regions are put into the same joined cluster. The weight $w_{k_1 k_2 \dots k_R}$ of each hyperedge is the sum of the costs of joining every pair of clusters in the hyperedge:

$$w_{k_1 k_2 \dots k_R} = \sum_{r_1=1}^{R-1} \sum_{r_2=r_1+1}^R C_{r_1 r_2 k_{r_1} k_{r_2}}$$

The weights are stacked as a column vector $\mathbf{w}$ with K^R rows (the number of hyperedges).

An incidence matrix $\mathbf{A}$ is then constructed with one row for each node N_{rk} and one column for each hyperedge $E_{k_1 k_2 \dots k_R}$. Each element is 1 if the hyperedge (column) contains the node (row), otherwise 0. Because its dimensions are RK rows by K^R columns, $\mathbf{A}$ can be very large, so it is implemented as a sparse `csr_matrix` in SciPy [11].

Then, the optimal combination of hyperedges $\mathbf{x}$ (a column vector with K^R rows) is selected using an integer linear program to minimize the total weight:

$$\min_{\mathbf{x} \in \{0,1\}^{K^R}} \mathbf{w}^T \mathbf{x}$$

subject to

$$\mathbf{A}\mathbf{x} = \mathbf{1}$$

which constrains every node (row of $\mathbf{A}$) to appear in exactly one hyperedge. The program is solved using the `milp` function from the `optimize` package of SciPy [11]. Because the joined clusters can be numbered arbitrarily, the convention is to copy the cluster numbering from the first region. Finally, the optimal way to match clusters from multiple regions is written into the report file.

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